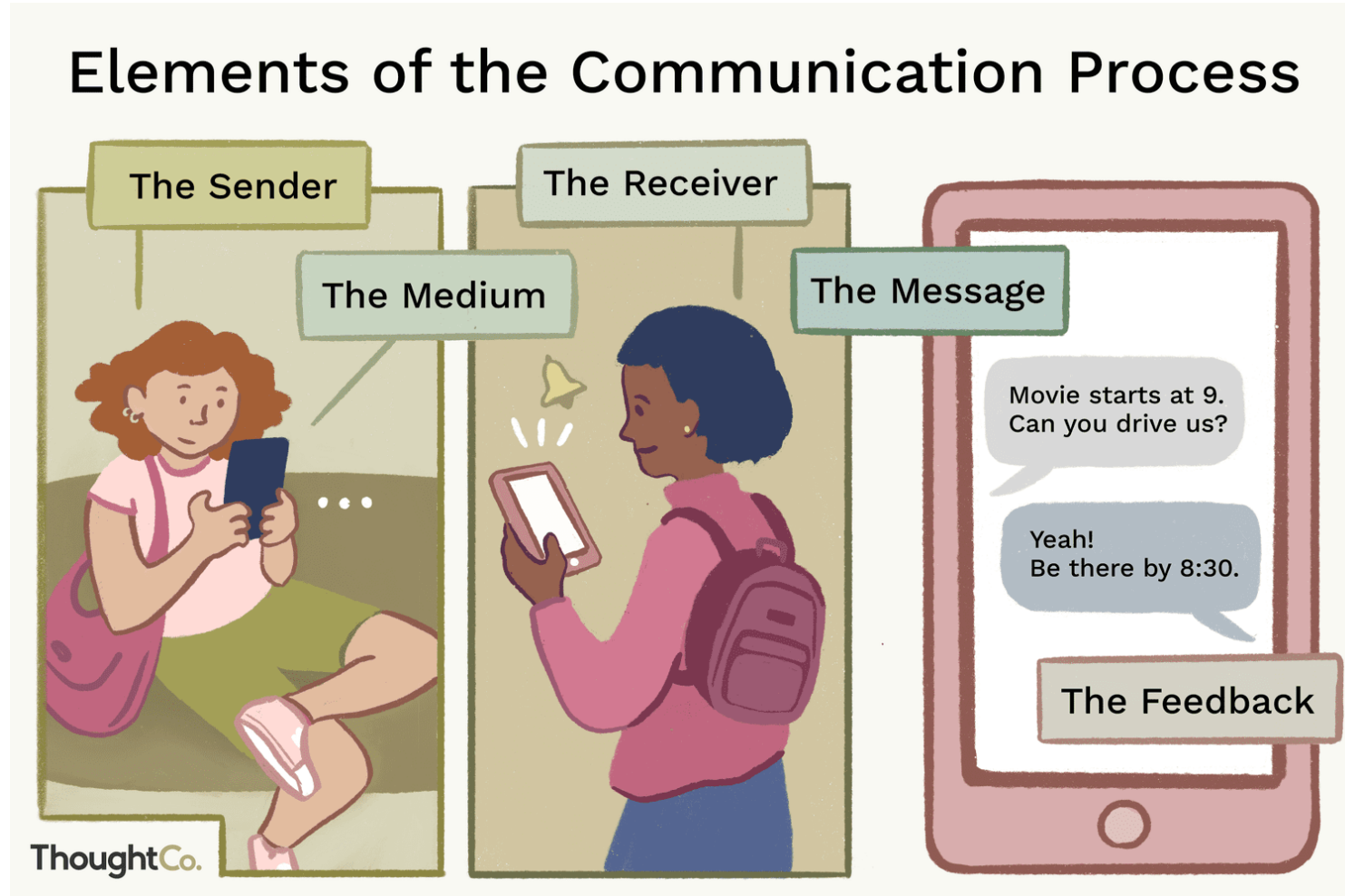


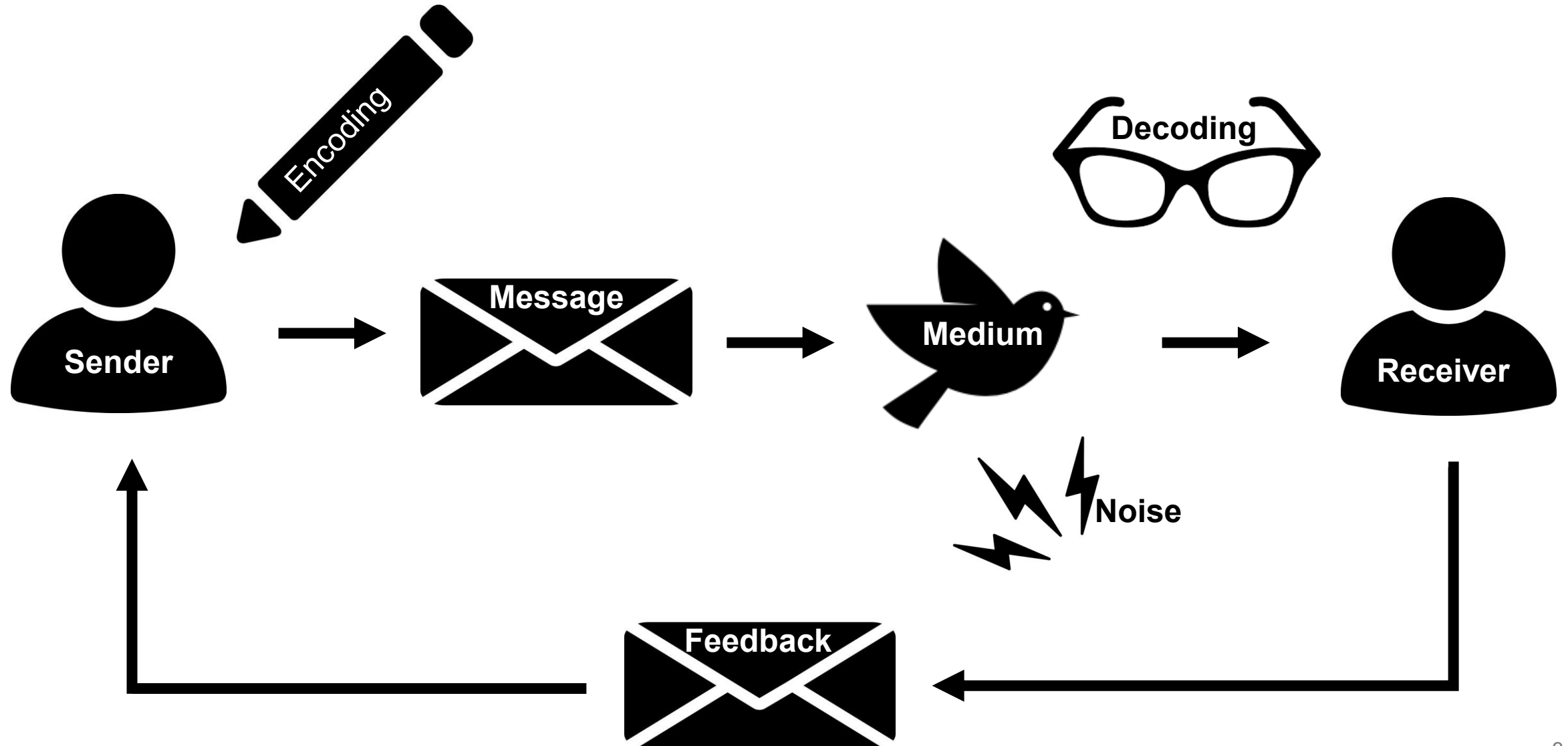
Network Fundamentals

Network protocols

What is communication



What is communication



Communication rules

Establishing the **Rules**

- An identified sender and receiver
- Agreed upon method of communicating (face-to-face, telephone, letter, photograph)
- Common language and grammar
- Encoding the messages for the medium
- Speed and timing of delivery
- Confirmation or acknowledgement requirements

Message formatting and encoding

Example: Personal letter contains the following elements:

- An identifier of the **recipient**
- The **message**:
 - A salutation or greeting
 - The message content
 - A closing phrase
- An identifier of the **sender**
- In addition to having the correct format, most personal letters must also be enclosed, or encapsulated, in an **envelope** for delivery



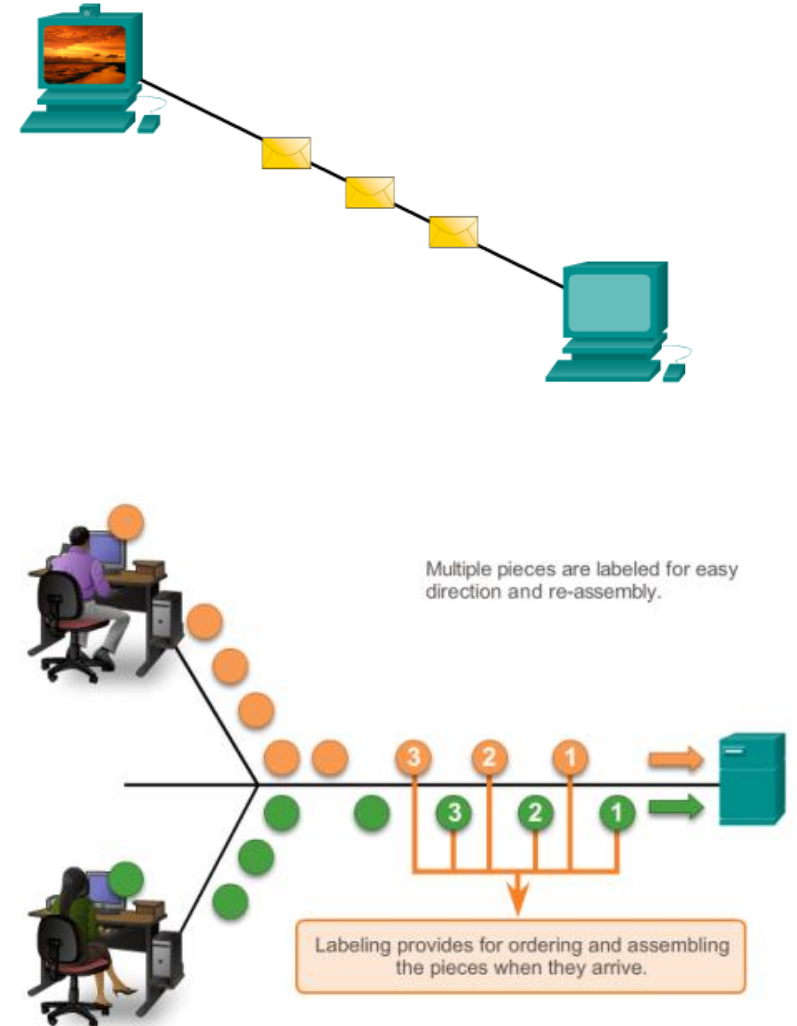
Message Size and segmentation

The **size restrictions of the message** require the source host to **break a long message into individual pieces** that meet both the minimum and maximum size requirements.

This is known as **segmenting**. Each segment is encapsulated in a separate message with the address information and is sent over the network.

At the receiving host, the messages are de-encapsulated and put back together to be processed and interpreted.

Segmenting allows multiple users or applications to use the same medium at the same time



Message Timing

Access Method

For example, you may be able to speak whenever you have something to say. Usually, a person must wait until no one else is talking before speaking. If two people talk at the same time, a **collision** of information occurs and it is necessary for the two to back off and start again

Flow Control

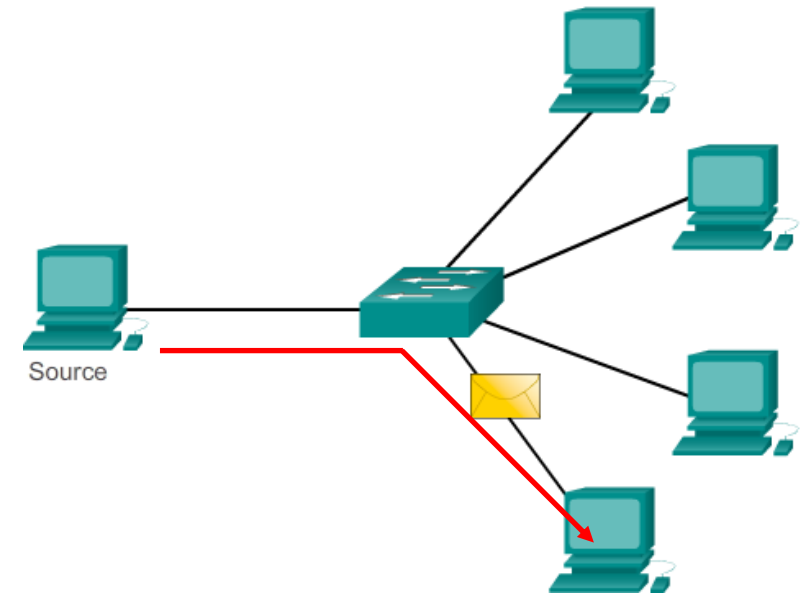
If one person speaks too quickly, it is difficult for the other person to hear and understand the message

Response Timeout

If a person asks a question and does not hear a response within an **acceptable amount of time**, the person assumes that no answer is coming and reacts with the same question

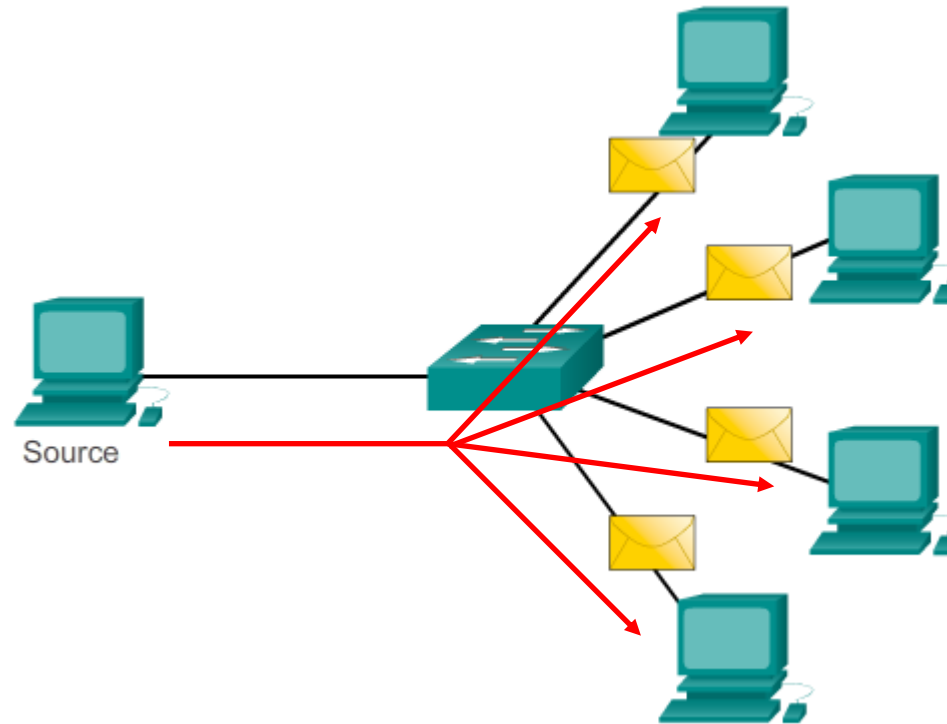
Message Delivery Options

Unicast: one to **one**



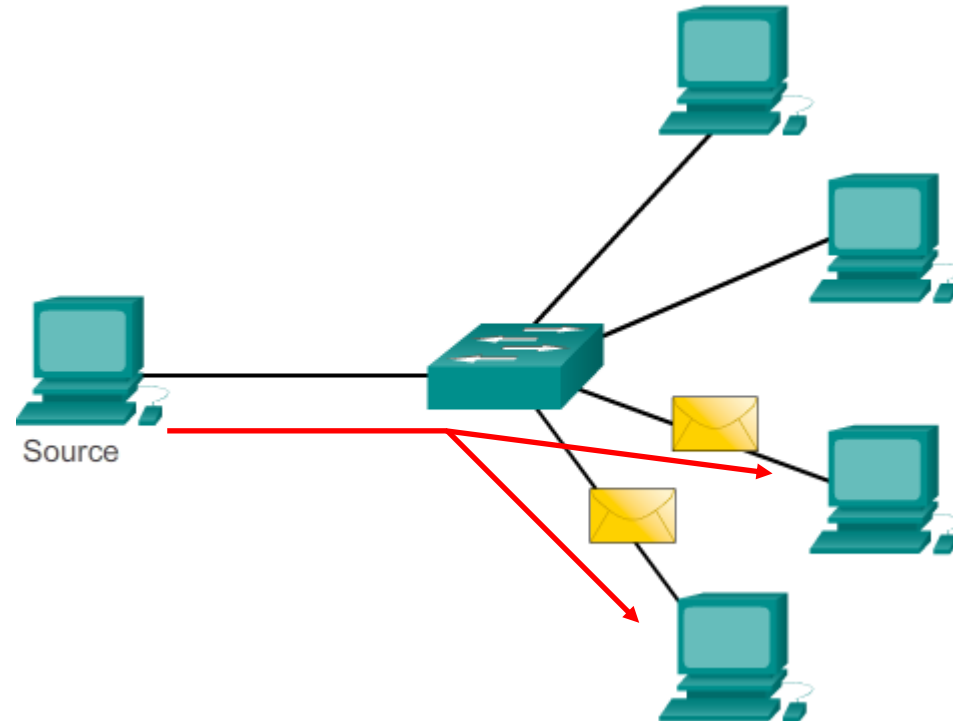
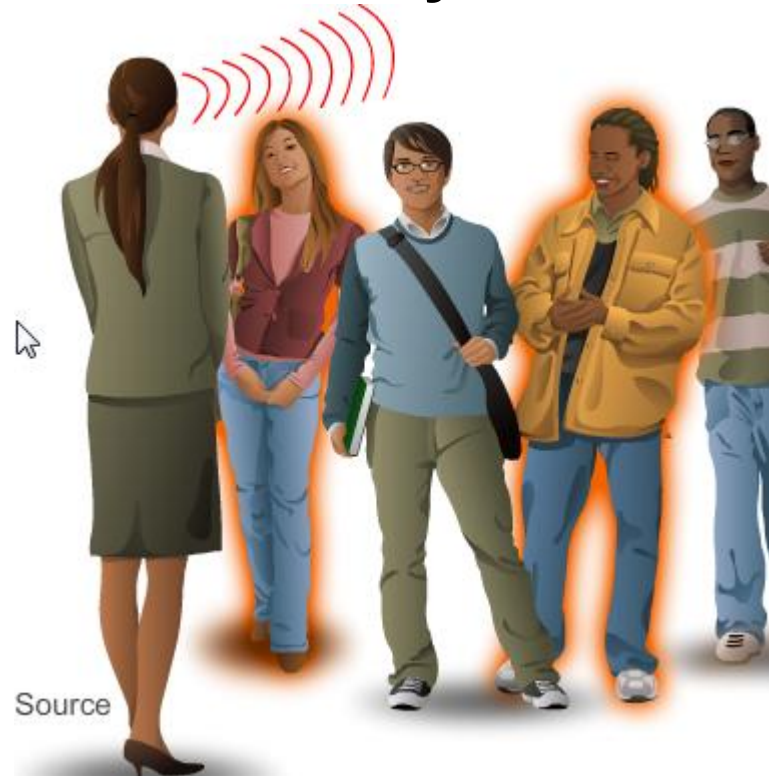
Message Delivery Options

Broadcast: one to **all**



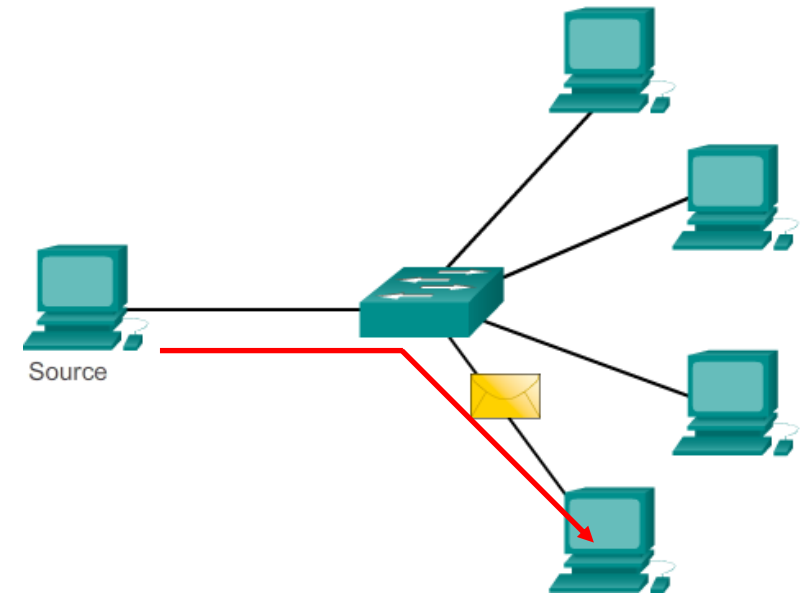
Message Delivery Options

Multicast: one to **many**



Message Delivery Options

Anycast: one to **closest**

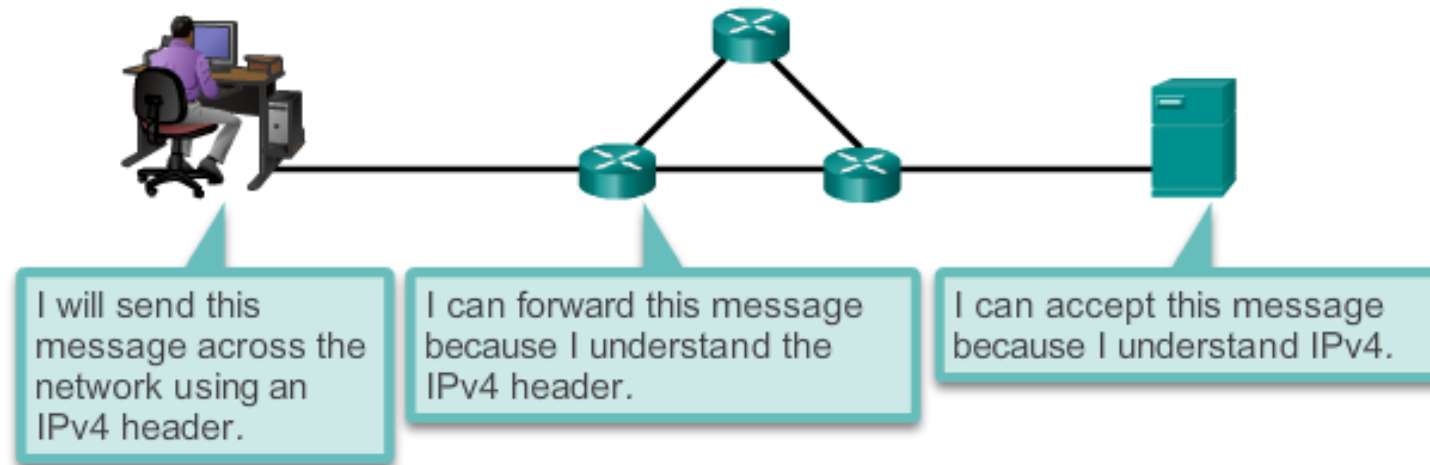


Network protocols

- A set of rules for formatting and processing data
- Each protocol has a specific, limited task
- Multiple protocols work together to make the entire process work
 - E.g. DNS gets the IP address of the webserver, HTTP gets the page from the webserver
- Protocols rely on other protocols
 - E.g. HTTP relies on IP to deliver the HTTP message to the webserver

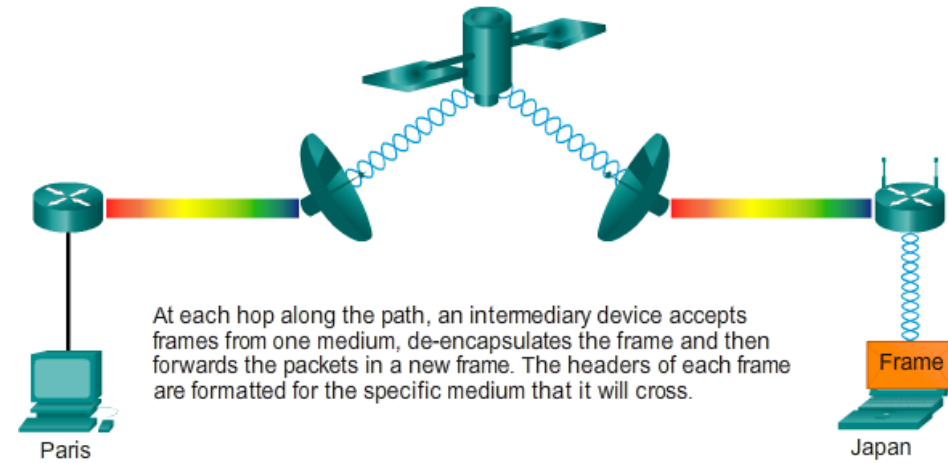
IP protocol

- Protocol for sending data packets on an IP network
- Address network devices with IP address
- Route packets through the IP network



Ethernet protocol

- Protocol for sending streams of bits over specific physical medium
- Ethernet uses physical addresses (MAC addresses) to transmit data on a local network
- An IP packet can travel over multiple types of ethernet connections
- Ethernet protocol is implemented in the network interface card or network devices



TCP and UDP protocols

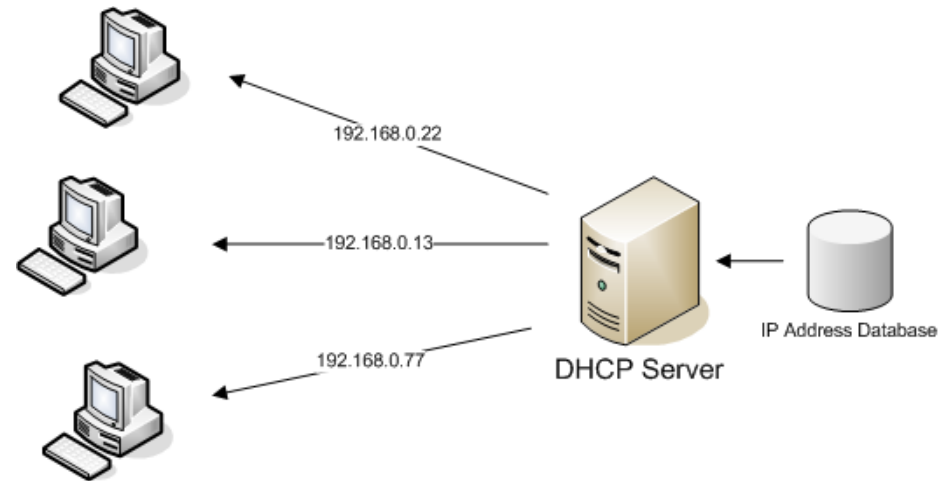
- Splitting application data into small packets
- Delivering the data to the correct application using port numbers
- TCP makes sure all packets arrive at the destination by acknowledging the received packets

ICMP: Internet Control Message Protocol

- Network administration and control
- ping

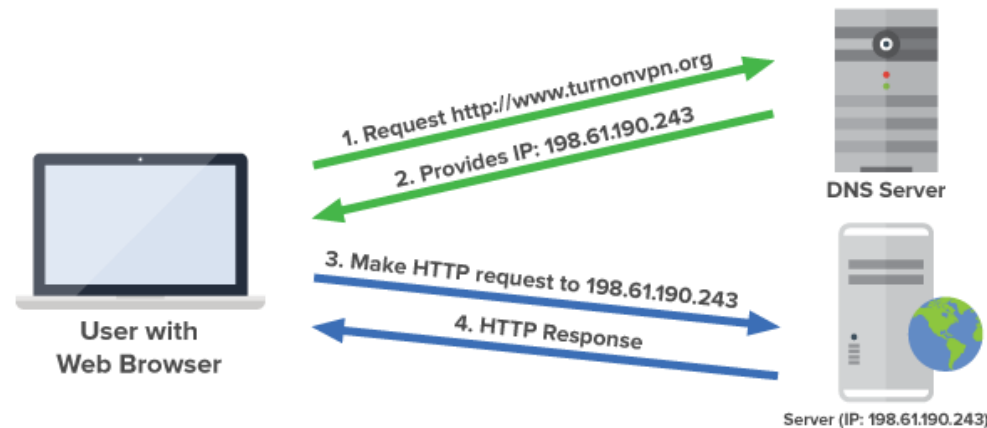
Network protocols - DHCP

- Dynamic Host Configuration Protocol
- A new connected computer needs an ip address
 - Static – IP address configured on client
 - DHCP – DHCP server provides IP address + options to devices which broadcast a DHCP request



Network protocols - DNS

- Domain Name System
- Translate domain name into IP address
- Local DNS – public DNS (8.8.8.8)
- Static – DHCP
- IP Address becomes destination address in IP header



Services (applications) on a network

- Receive an IP address in the network DHCP
- Translate domain name to IP address DNS
- Get time NTP
- Webserver HTTP,HTTPS
- Fileserver FTP,TFTP

Layered model

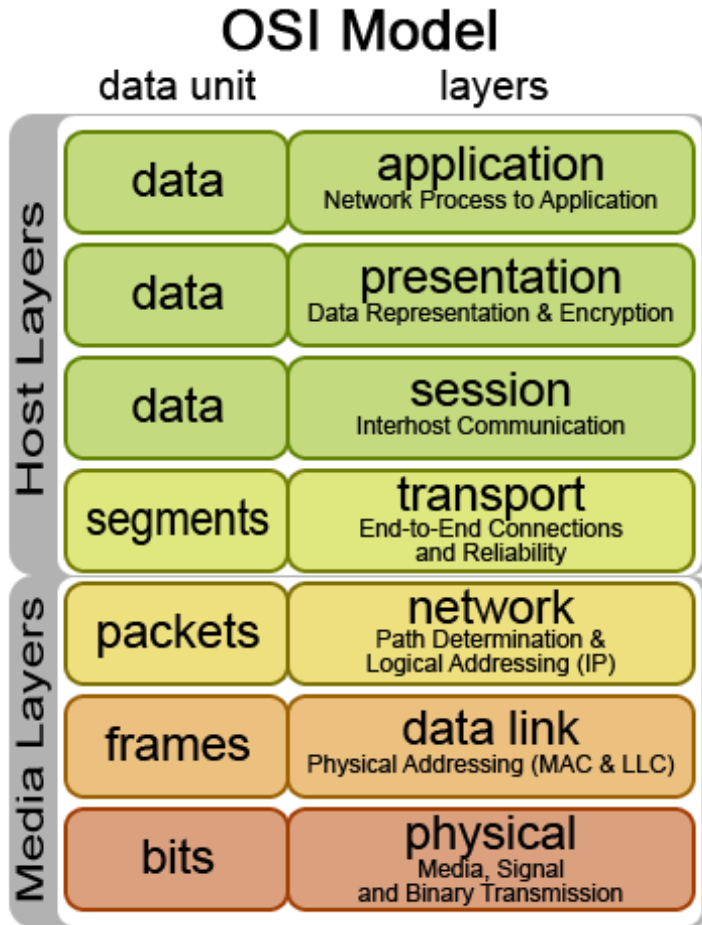
Protocols with a similar function are **grouped within a layer**.

Protocols can **interact** with protocols from a **layer above and below it**.

Benefits

- Assists in protocol design, because protocols that operate at a specific layer have **defined tasks**. They can focus on that specific task and can rely on other protocols for other tasks.
- **Prevents** technology or capability changes in one layer from **affecting other layers** above and below.
- Fosters competition because products from **different vendors** can work together. No Vendor lock in.
- Provides a common language to describe networking functions and capabilities.
- Makes **troubleshooting** easier

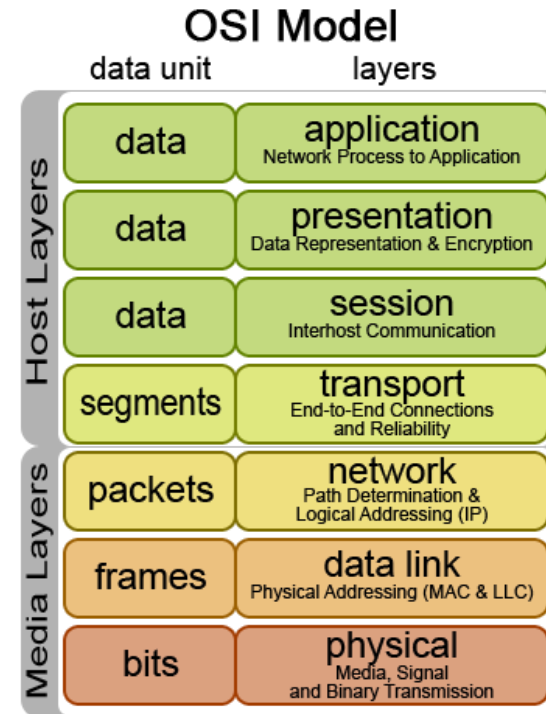
The OSI Reference Model



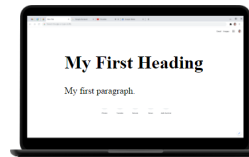
- The **application layer** provides the means for end-to-end connectivity between individuals in the human network using data networks.
- The **presentation layer** provides for common representation of the data transferred between application layer services.
- The **session layer** provides services to the presentation layer to organize its dialogue (dialog control) to manage data exchange.
- The **transport layer** defines services to segment, transfer, and reassemble the data for individual communications between the applications on end devices.
- The **network layer** provides services to exchange the individual pieces of data over the network between identified end devices.
- The **data link layer** protocols describe methods for exchanging data frames between devices over a common media.
- The **physical layer** protocols describe the mechanical, electrical, functional, maintain, and de-activate physical-connections for bit transmission to and from a network device.

The OSI model – Webserver example

- Application Protocol – Hypertext Transfer Protocol (HTTP)
- Transport Protocol – Transmission Control Protocol (TCP) called **segments**, manages the individual conversations between web servers and web clients
- Network Protocol – Internet Protocol (IP), encapsulate segments into **packets** and assigning addresses
- Data link Protocols - Ethernet
 - Data link layer, format IP packet to be transmit over the media
 - physical layer, transmission of data on the network media.



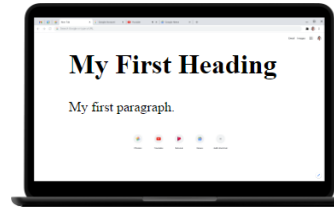
NGINX



Layers of the osi model and their implementation



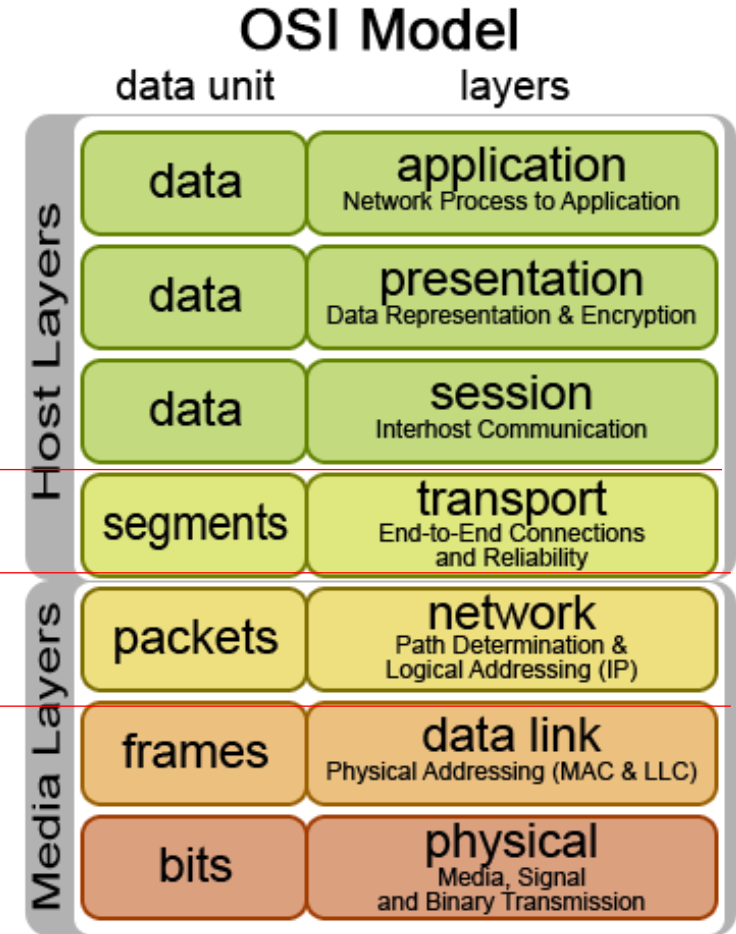
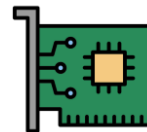
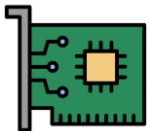
NGINX



Linux



Windows



Protocol Suites and Industry Standards

A protocol suite is a **set of protocols** that work together to provide comprehensive network communication services

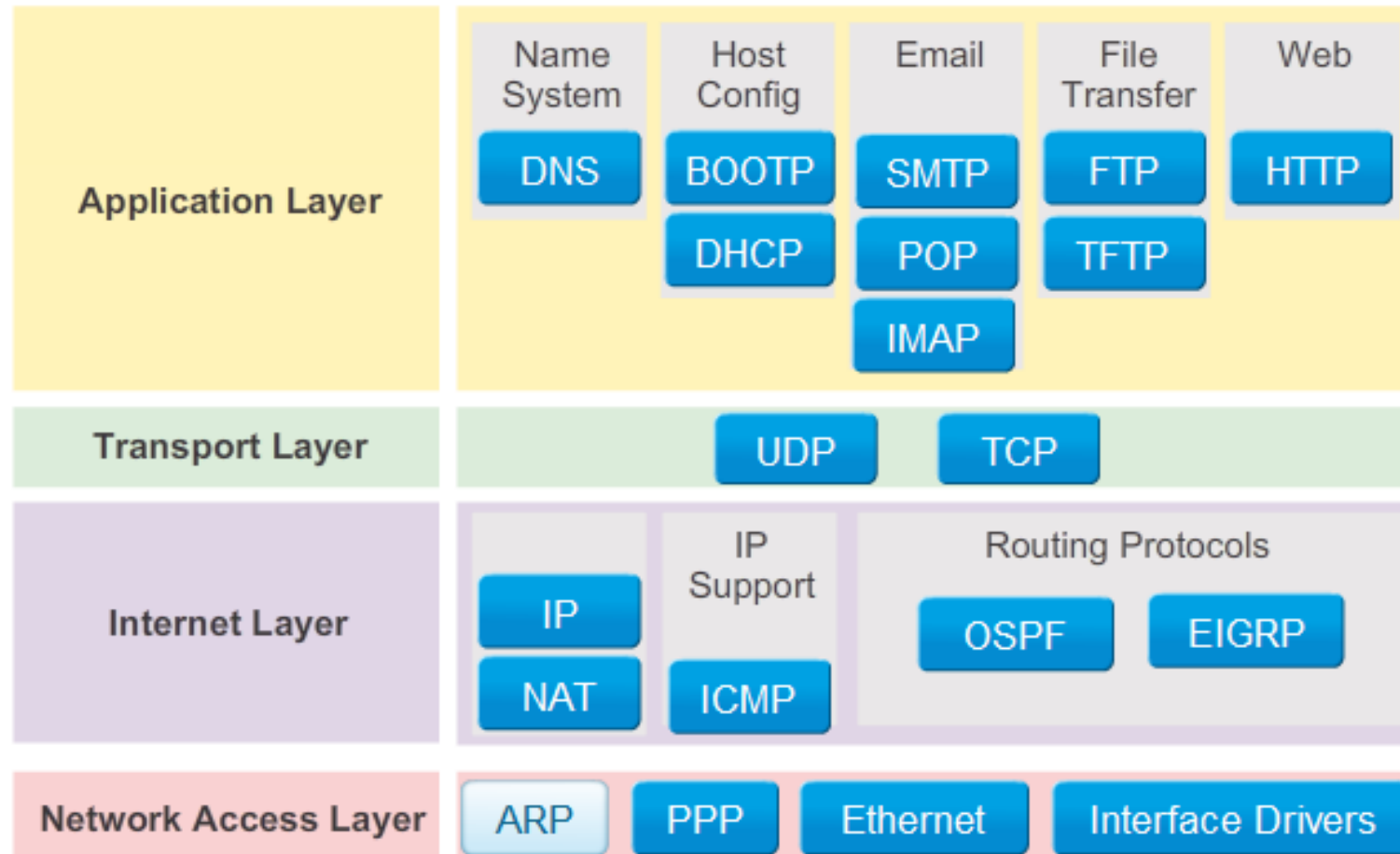
The open standards-based **TCP/IP** has replaced other vendor proprietary protocol suites, such as:

- Apple's AppleTalk
- Novell Netware IPX/SPX

	TCP/IP	ISO	AppleTalk	Novell Netware
7 6 5	HTTP DNS DHCP FTP	ACSE ROSE TRSE SESE	AFP	NDS
4	TCP UDP	TP0 TP1 TP2 TP3 TP4	ATP AEP NBP RTMP	SPX
3	IPv4 IPv6 ICMPV4 ICMPV6	CONP/CMNS CLNP/CLNS	AFP	IPX
2 1	Ethernet PPP Frame Relay ATM WLAN			

TCP/IP Protocol Suite and Communication

TCP/IP Protocol Suite and Communication Process



TCP/IP Application layer

DNS: Translates domain names, such as cisco.com, into IP addresses

DHCP: Dynamically assigns IP addresses to client stations at start-up

SMTP: Enables clients to send email to a mail server

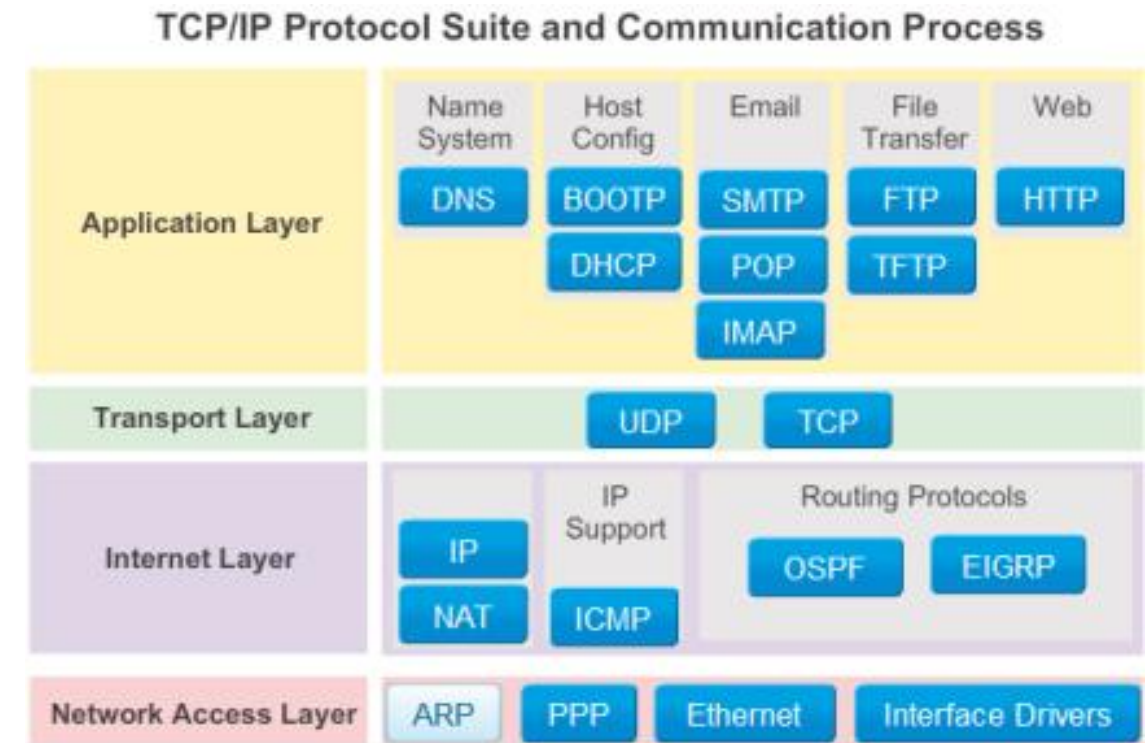
POP: Post office protocol .Enables clients to retrieve email from a mail server

IMAP: Internet message access protocol. Enables clients to access email stored on a mail server

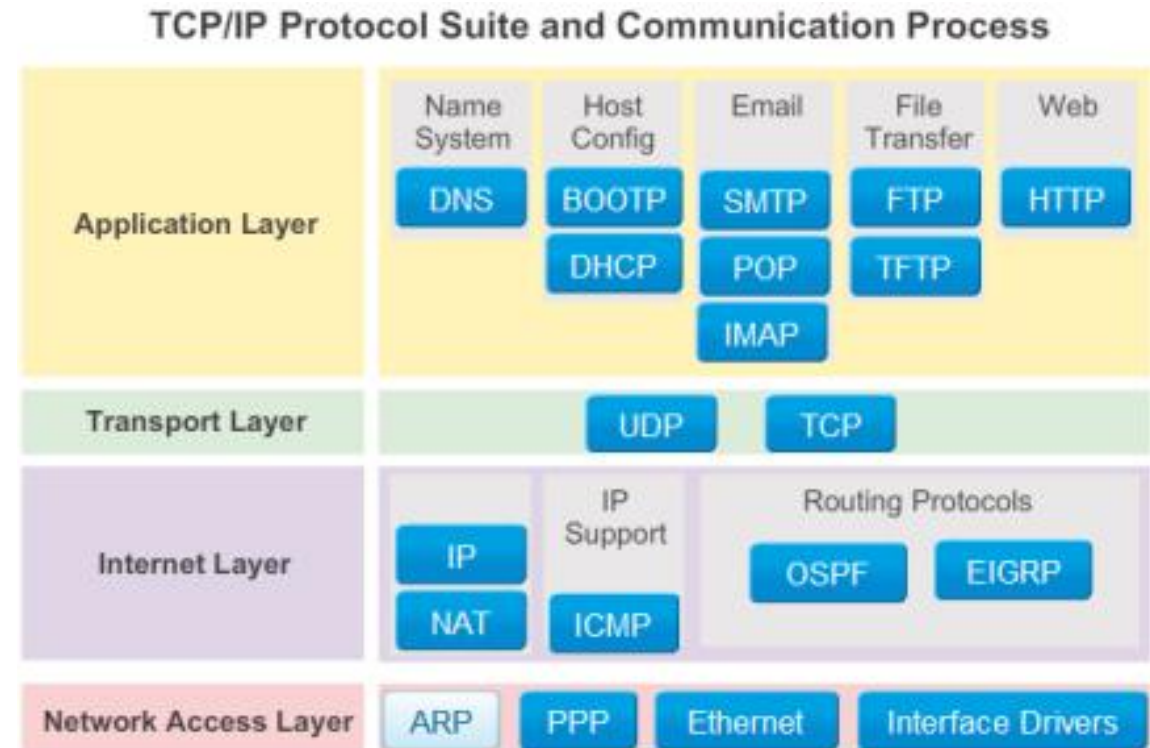
FTP: access and transfer files to and from another host over a network

TFTP: A simple, connectionless file transfer protocol

HTTP: Set of rules for exchanging text, graphic images, sound, video, and other multimedia files on the World Wide Web



- **UDP:** user datagram protocol
a process running on one host to send packets to a process running on another host. Does not confirm successful datagram transmission
- **TCP:** transmission control protocol
Enables reliable communication between processes running on separate hosts
Reliable, acknowledged transmissions that confirm successful delivery
- **IP:** internet protocol
Packages messages into packets
Addresses packets for end-to-end delivery over an Internetwork
- **NAT:** network address translation
Translates IP addresses from a private network into globally unique public IP addresses
- **ICMP:** internet control message protocol
Provides feedback from a destination host to a source host about errors in packet delivery (ping)



OSPF: open shortest path first
Link-state routing protocol

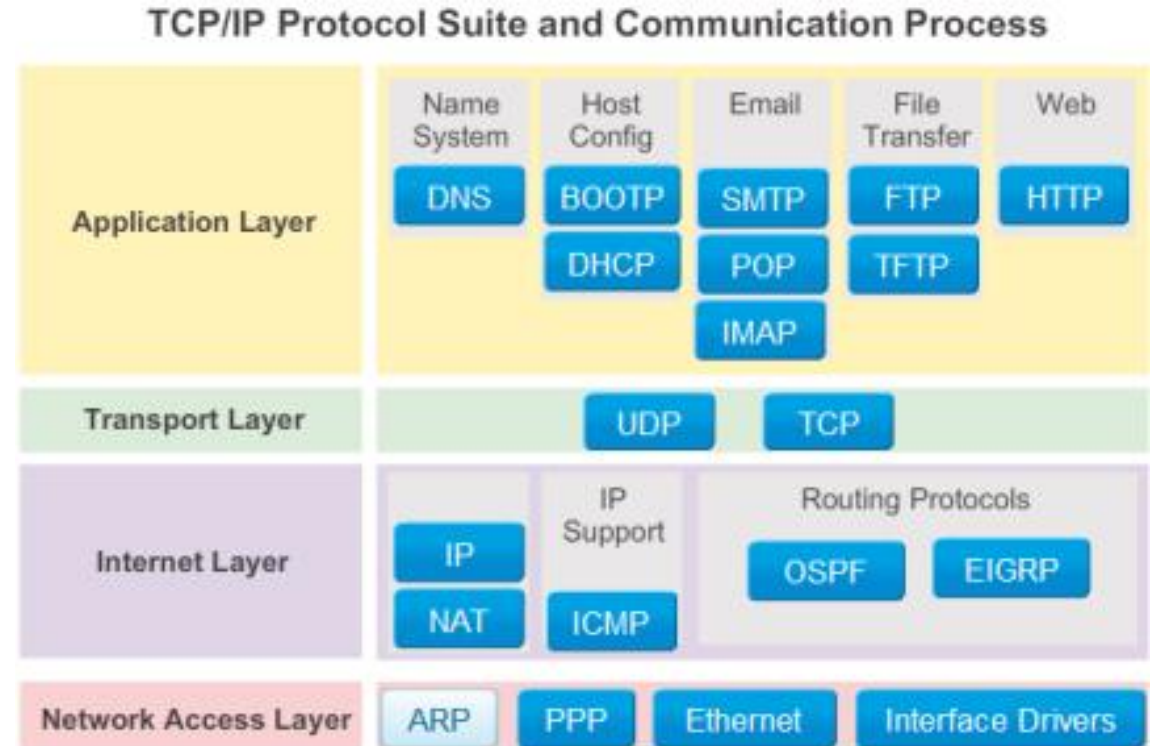
EIGRP: enhanced interior gateway routing
protocol. Cisco proprietary routing protocol

ARP: address resolution protocol
I have an IP address, give me your
hardware (MAC) address

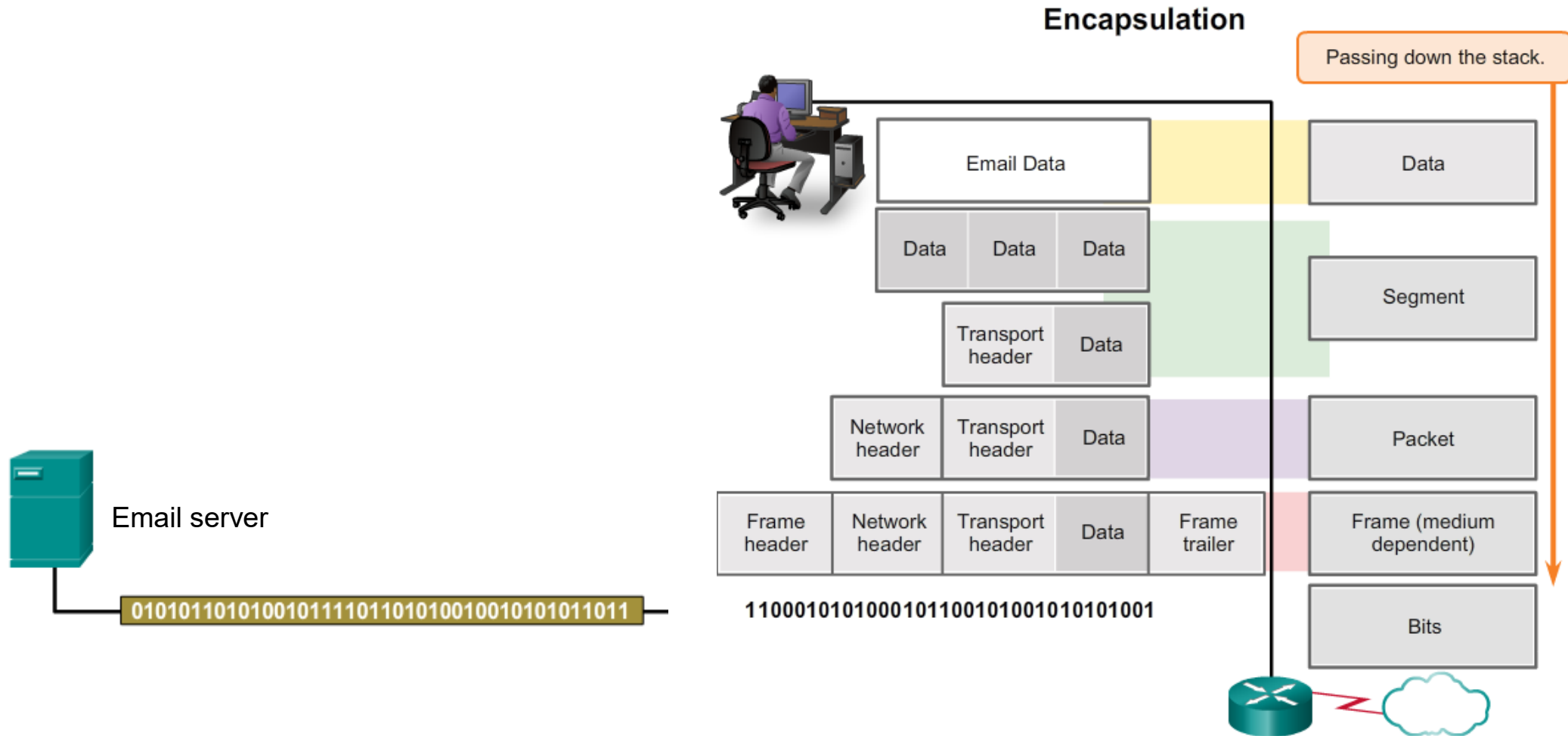
PPP: point to point protocol
Provides a means of encapsulating packets
for transmission over a serial link

Ethernet: Defines the rules for wiring and
signalling standards of the network access
layer

Interface drivers: Provides instruction to a
machine for the control of a specific
interface on a network device

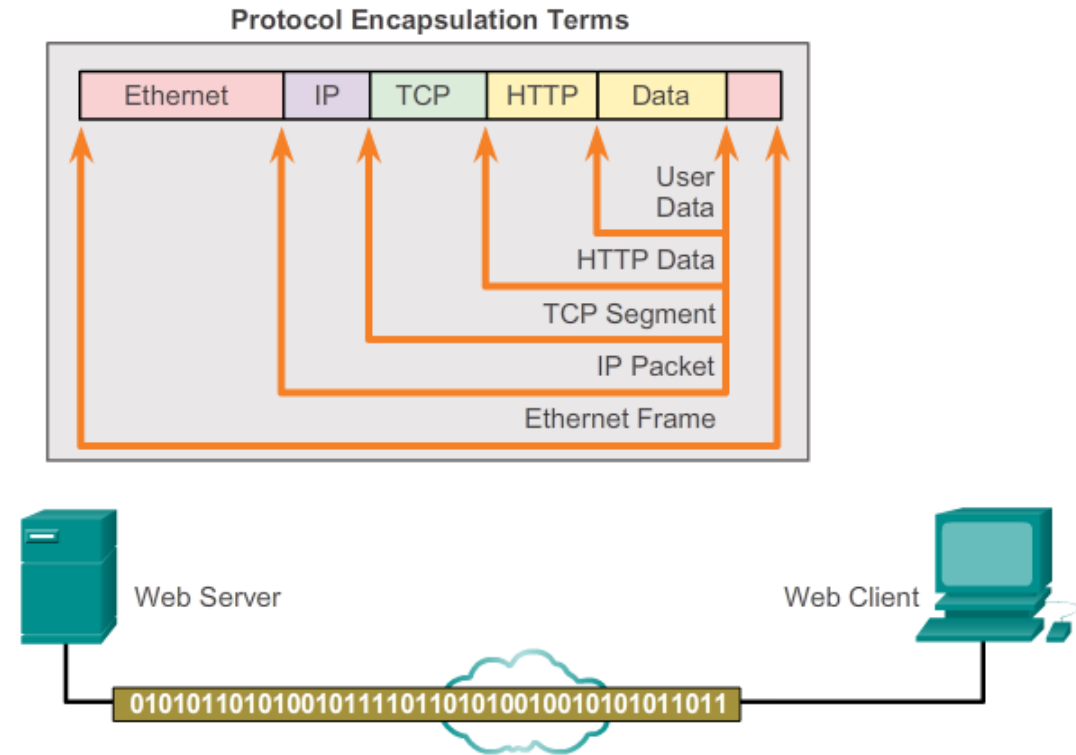


Protocol Data Units (PDUs)

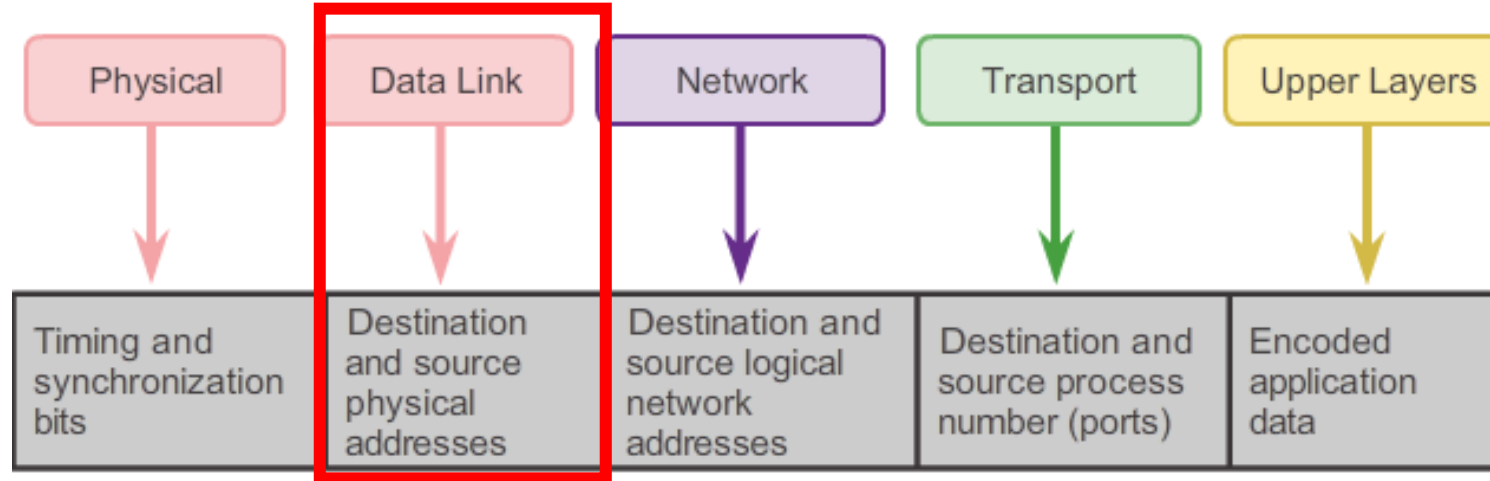


Headers – Webserver example

1. The web server's Hypertext Markup Language (HTML) page is the **data** to be sent.
2. HTTP header is added to the front of the HTML data.
3. The TCP transport layer protocol is used to manage the individual conversation between the web server and web client. **(segment)**
4. IP assigns the appropriate source and destination IP addresses. This information is known as an **IP packet**.
5. The Ethernet protocol adds information to both ends of the IP packet, known as a data link **frame**.
6. This data is now transported through the internetwork, which consists of media and intermediary devices. **(bits)**
7. The client receives the data link frames
8. The web page information is then passed on to the client's web browser software.



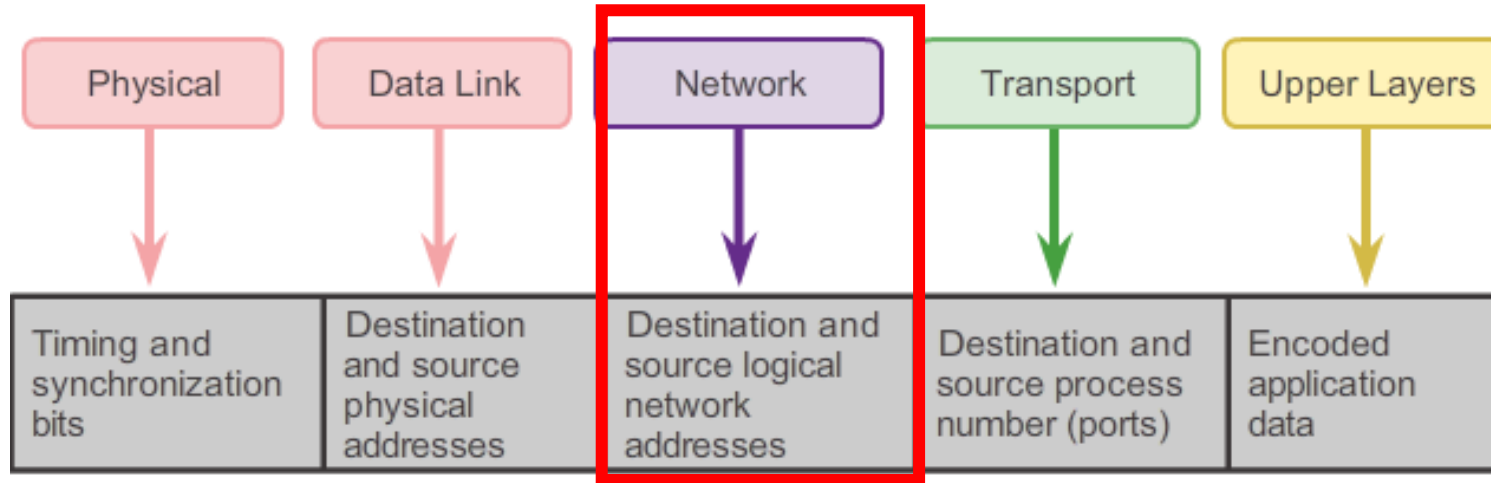
Headers - Datalink



Data Link Address

- **Source MAC address** - The physical MAC address of the device that is sending the packet.
- **Destination MAC address** - The physical MAC address of the network interface of either the next hop router or the network interface of the destination device.
- **Addresses used on the local network**

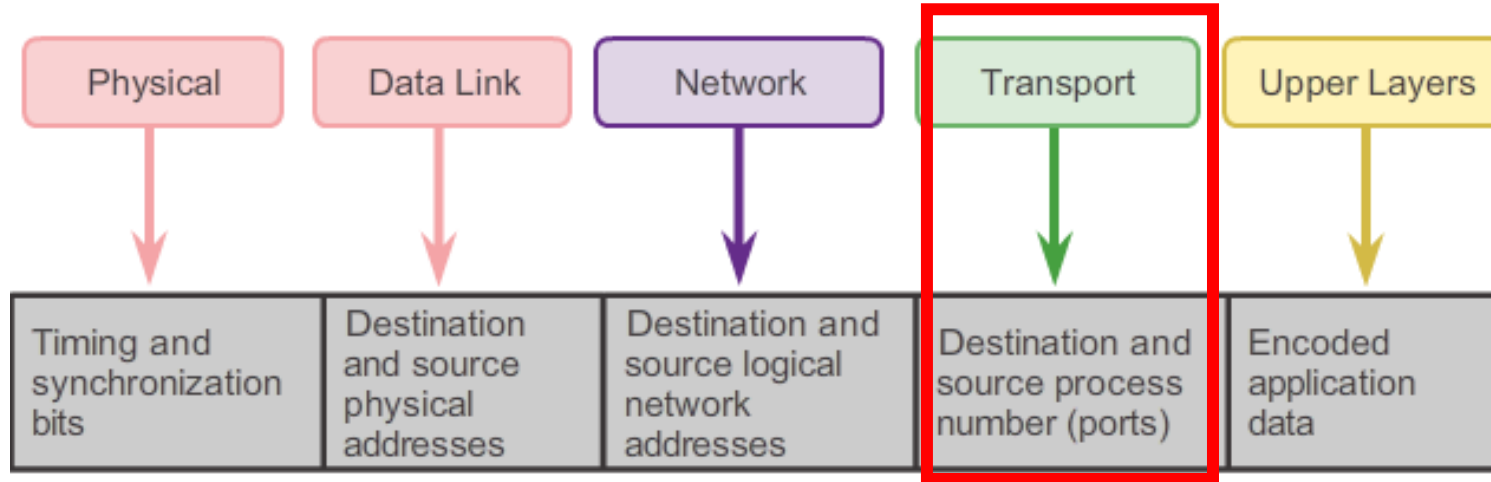
Headers - Network



An IP packet contains two IP addresses:

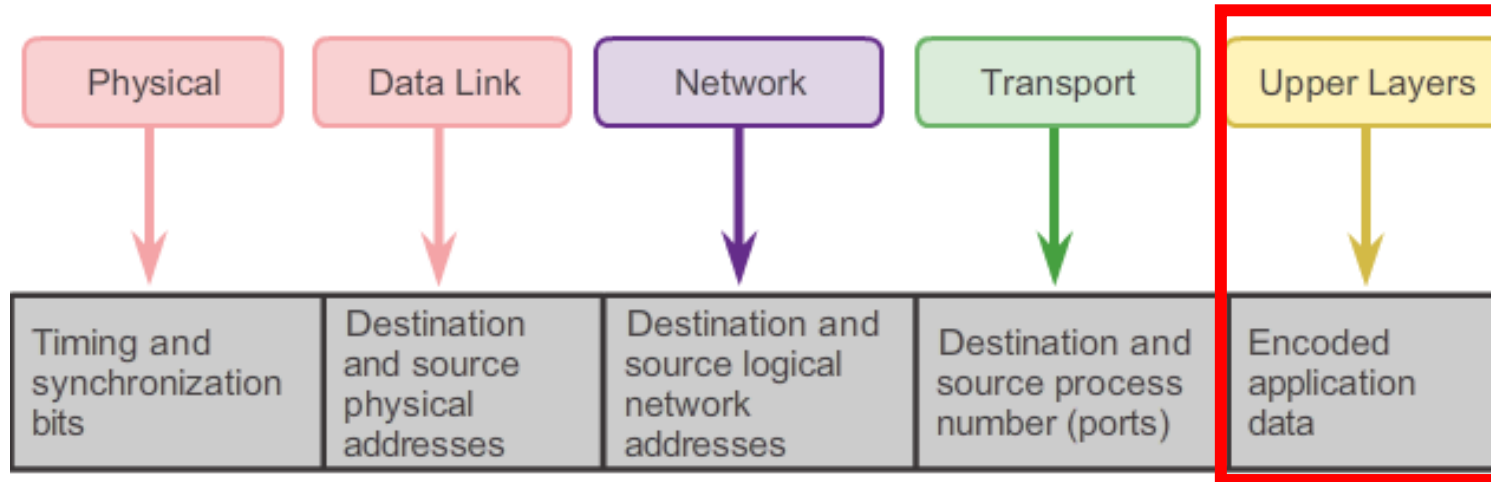
- **Source IP address** - The IP address of the sending device.
- **Destination IP address** - The IP address of the receiving device. The destination IP address is used by routers to forward a packet to its destination.
- **End to end addresses**

Headers - Transport



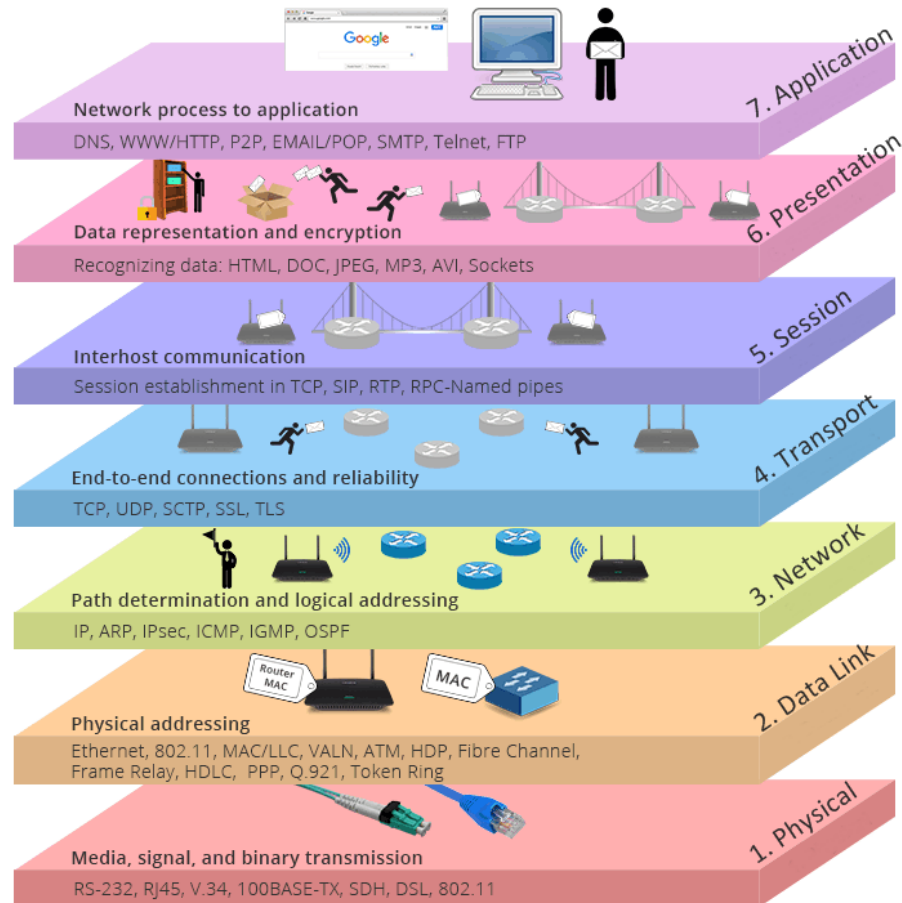
- An TCP/UDP packet contains two **port numbers**:
- **Source Port** – Identifies the application on the sender's device
- **Destination Port** - Identifies the application on the receiver's
- Sequence numbers to put the frames in the correct order

Headers - Application



- Application specific data
- Formatted to the application protocol

Understand the OSI model



Wireshark - Ethernet

http						
No.	Time	Source	Destination	Protocol	Length	Info
80	2.745446	10.0.14.123	146.190.62...	HTTP	500	GET / HTTP/1.1
83	2.933884	146.190.62...	10.0.14.123	HTTP	1350	HTTP/1.1 200 OK (text/html)

- Frame 83: 1350 bytes on wire (10800 bits), 1350 bytes captured (10800 bits) on interface \Device\NPF_{...}
- Ethernet II, Src: VMware_fb:a0:c9 (00:0c:29:fb:a0:c9), Dst: Intel_9d:37:e9 (38:87:d5:9d:37:e9)
 - Destination: Intel_9d:37:e9 (38:87:d5:9d:37:e9)
 - Source: VMware_fb:a0:c9 (00:0c:29:fb:a0:c9)
 - Type: IPv4 (0x0800)
 - [Stream index: 0]
- Internet Protocol Version 4, Src: 146.190.62.39, Dst: 10.0.14.123
- Transmission Control Protocol, Src Port: 80, Dst Port: 9180, Seq: 1461, Ack: 447, Len: 1296
- [2 Reassembled TCP Segments (2756 bytes): #82(1460), #83(1296)]
- Hypertext Transfer Protocol, has 2 chunks (including last chunk)
- Line-based text data: text/html (100 lines)

Wireshark - IP

http						
No.	Time	Source	Destination	Protocol	Length	Info
80	2.745446	10.0.14.123	146.190.62...	HTTP	500	GET / HTTP/1.1
83	2.933884	146.190.62...	10.0.14.123	HTTP	1350	HTTP/1.1 200 OK (text/html)

- > Frame 83: 1350 bytes on wire (10800 bits), 1350 bytes captured (10800 bits) on interface \Device\NPF_{...}
- > Ethernet II, Src: VMware_fb:a0:c9 (00:0c:29:fb:a0:c9), Dst: Intel_9d:37:e9 (38:87:d5:9d:37:e9)
- ✓ Internet Protocol Version 4, Src: 146.190.62.39, Dst: 10.0.14.123
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 1336
 - Identification: 0x65f7 (26103)
 - > 010. = Flags: 0x2, Don't fragment
 - ...0 0000 0000 0000 = Fragment Offset: 0
 - Time to Live: 46
 - Protocol: TCP (6)
 - Header Checksum: 0xf868 [validation disabled]
[Header checksum status: Unverified]
 - Source Address: 146.190.62.39
 - Destination Address: 10.0.14.123
 - [Stream index: 12]
- > Transmission Control Protocol, Src Port: 80, Dst Port: 9180, Seq: 1461, Ack: 447, Len: 1296
- > [2 Reassembled TCP Segments (2756 bytes): #82(1460), #83(1296)]
- > Hypertext Transfer Protocol, has 2 chunks (including last chunk)
- > Line-based text data: text/html (100 lines)

Wireshark - TCP

http						
No.	Time	Source	Destination	Protocol	Length	Info
→ 80	2.745446	10.0.14.123	146.190.62...	HTTP	500	GET / HTTP/1.1
← 83	2.933884	146.190.62...	10.0.14.123	HTTP	1350	HTTP/1.1 200 OK (text/html)

>	Frame 83: 1350 bytes on wire (10800 bits), 1350 bytes captured (10800 bits) on interface \Device\NPF_...
>	Ethernet II, Src: VMware_fb:a0:c9 (00:0c:29:fb:a0:c9), Dst: Intel_9d:37:e9 (38:87:d5:9d:37:e9)
>	Internet Protocol Version 4, Src: 146.190.62.39, Dst: 10.0.14.123
√	Transmission Control Protocol, Src Port: 80, Dst Port: 9180, Seq: 1461, Ack: 447, Len: 1296
	Source Port: 80
	Destination Port: 9180
	[Stream index: 11]
	[Stream Packet Number: 7]
>	[Conversation completeness: Incomplete, DATA (15)]
	[TCP Segment Len: 1296]
	Sequence Number: 1461 (relative sequence number)
	Sequence Number (raw): 3381052579
	[Next Sequence Number: 2757 (relative sequence number)]
	Acknowledgment Number: 447 (relative ack number)
	Acknowledgment number (raw): 1171148023
	0101 = Header Length: 20 bytes (5)
>	Flags: 0x018 (PSH, ACK)
	Window: 501
	[Calculated window size: 64128]

Wireshark - HTTP

http						
No.	Time	Source	Destination	Protocol	Length	Info
80	2.745446	10.0.14.123	146.190.62...	HTTP	500	GET / HTTP/1.1
83	2.933884	146.190.62...	10.0.14.123	HTTP	1350	HTTP/1.1 200 OK (text/html)

>	Frame 83: 1350 bytes on wire (10800 bits), 1350 bytes captured (10800 bits) on interface \Device\NPF_...
>	Ethernet II, Src: VMware_fb:a0:c9 (00:0c:29:fb:a0:c9), Dst: Intel_9d:37:e9 (38:87:d5:9d:37:e9)
>	Internet Protocol Version 4, Src: 146.190.62.39, Dst: 10.0.14.123
>	Transmission Control Protocol, Src Port: 80, Dst Port: 9180, Seq: 1461, Ack: 447, Len: 1296
>	[2 Reassembled TCP Segments (2756 bytes): #82(1460), #83(1296)]
>	Hypertext Transfer Protocol, has 2 chunks (including last chunk)
>	HTTP/1.1 200 OK\r\n
>	Server: nginx/1.18.0 (Ubuntu)\r\n
>	Date: Sun, 15 Feb 2026 22:18:03 GMT\r\n
>	Content-Type: text/html\r\n
>	Last-Modified: Wed, 22 Mar 2023 14:54:48 GMT\r\n
>	Transfer-Encoding: chunked\r\n
>	Connection: keep-alive\r\n
>	ETag: W/"641b16b8-1404"\r\n
>	Referrer-Policy: strict-origin-when-cross-origin\r\n
>	X-Content-Type-Options: nosniff\r\n
>	Feature-Policy: accelerometer 'none'; camera 'none'; geolocation 'none'; gyroscope 'none'; magnetometer 'none';
>	[...]Content-Security-Policy: default-src 'self'; script-src cdnjs.cloudflare.com 'self' 'report-sha256';
>	Content-Encoding: gzip\r\n
>	\r\n

Wireshark - Data

http						
No.	Time	Source	Destination	Protocol	Length	Info
80	2.745446	10.0.14.123	146.190.62...	HTTP	500	GET / HTTP/1.1
83	2.933884	146.190.62...	10.0.14.123	HTTP	1350	HTTP/1.1 200 OK (text/html)

> Frame 83: 1350 bytes on wire (10800 bits), 1350 bytes captured (10800 bits) on interface \Device\NPF...
> Ethernet II, Src: VMware_fb:a0:c9 (00:0c:29:fb:a0:c9), Dst: Intel_9d:37:e9 (38:87:d5:9d:37:e9)
> Internet Protocol Version 4, Src: 146.190.62.39, Dst: 10.0.14.123
> Transmission Control Protocol, Src Port: 80, Dst Port: 9180, Seq: 1461, Ack: 447, Len: 1296
> [2 Reassembled TCP Segments (2756 bytes): #82(1460), #83(1296)]
> Hypertext Transfer Protocol, has 2 chunks (including last chunk)
v Line-based text data: text/html (100 lines)
 <!DOCTYPE HTML>\r\n
 <html>\r\n
 <head>\r\n
 <title>HTTP Forever</title>\r\n
 <meta http-equiv="content-type" content="text/html; charset=utf-8" />\r\n
 <meta name="description" content="A site that will always be available over HTTP!" />\r\n
 <meta name="keywords" content="HTTP WiFi Captive Portal" />\r\n
 <!--[if lte IE 8]><script src="css/ie/html5shiv.js"></script><![endif]-->\r\n
 <script src="https://cdnjs.cloudflare.com/ajax/libs/jquery/3.3.1/jquery.min.js" integrity="sha2...></script>\r\n
 <script src="https://cdnjs.cloudflare.com/ajax/libs/skel/3.0.1/skel.min.js" integrity="sha256-3...></script>\r\n
 <script src="https://cdnjs.cloudflare.com/ajax/libs/skel-layers/2.2.1/skel.min.js" integrity="s...></script>\r\n
 <script src="js/init.min.js"></script>\r\n
 <noscript>\r\n
 <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/skel-layers/2.2.1/skel.mi...>\r\n
 <link rel="stylesheet" href="css/style.min.css" />\r\n
 <link rel="stylesheet" href="css/style-wide.min.css" />\r\n
 </noscript>\r\n
 <!--[if lte IE 8]><link rel="stylesheet" href="css/ie/v8.min.css" /><![endif]-->\r\n